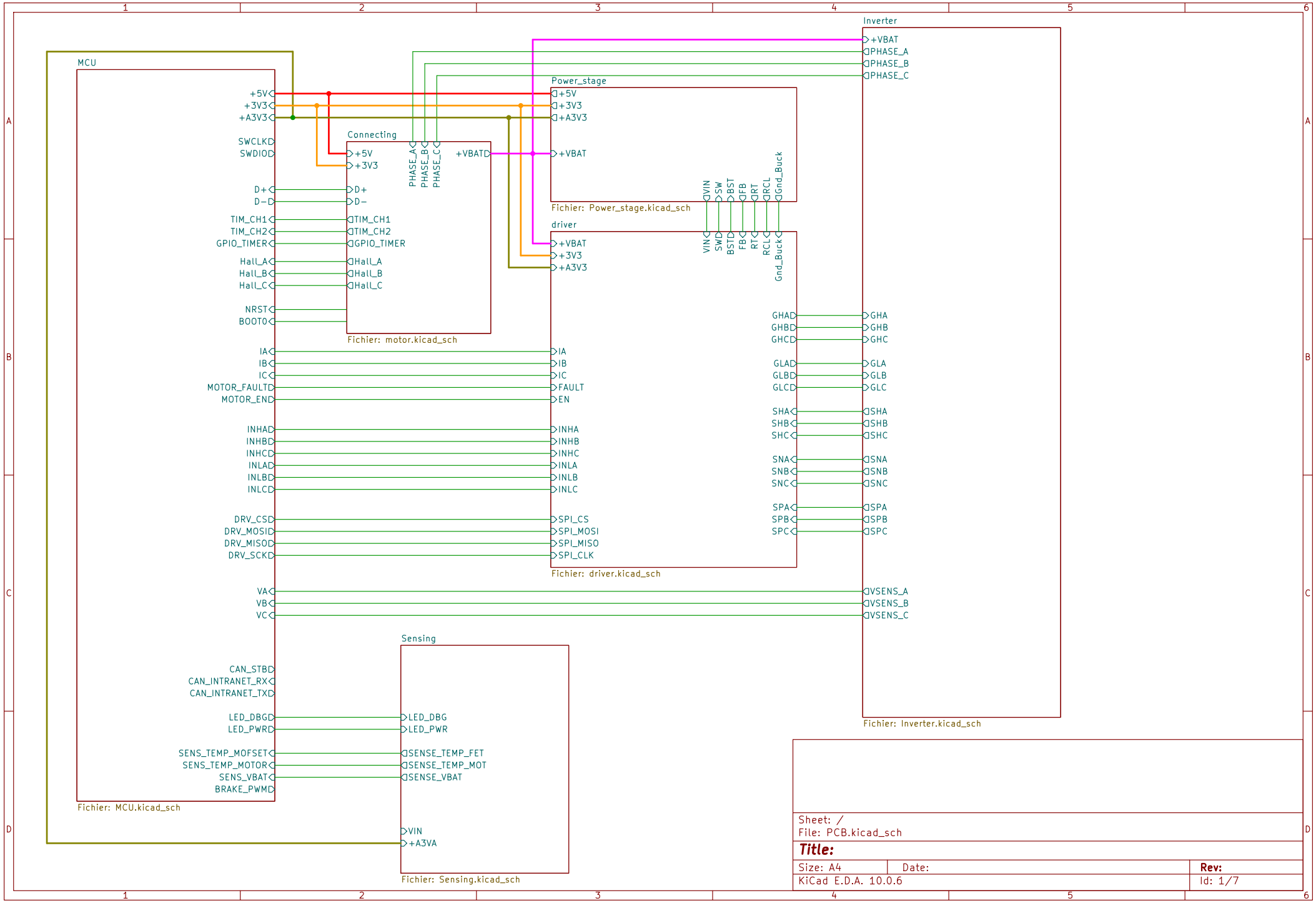
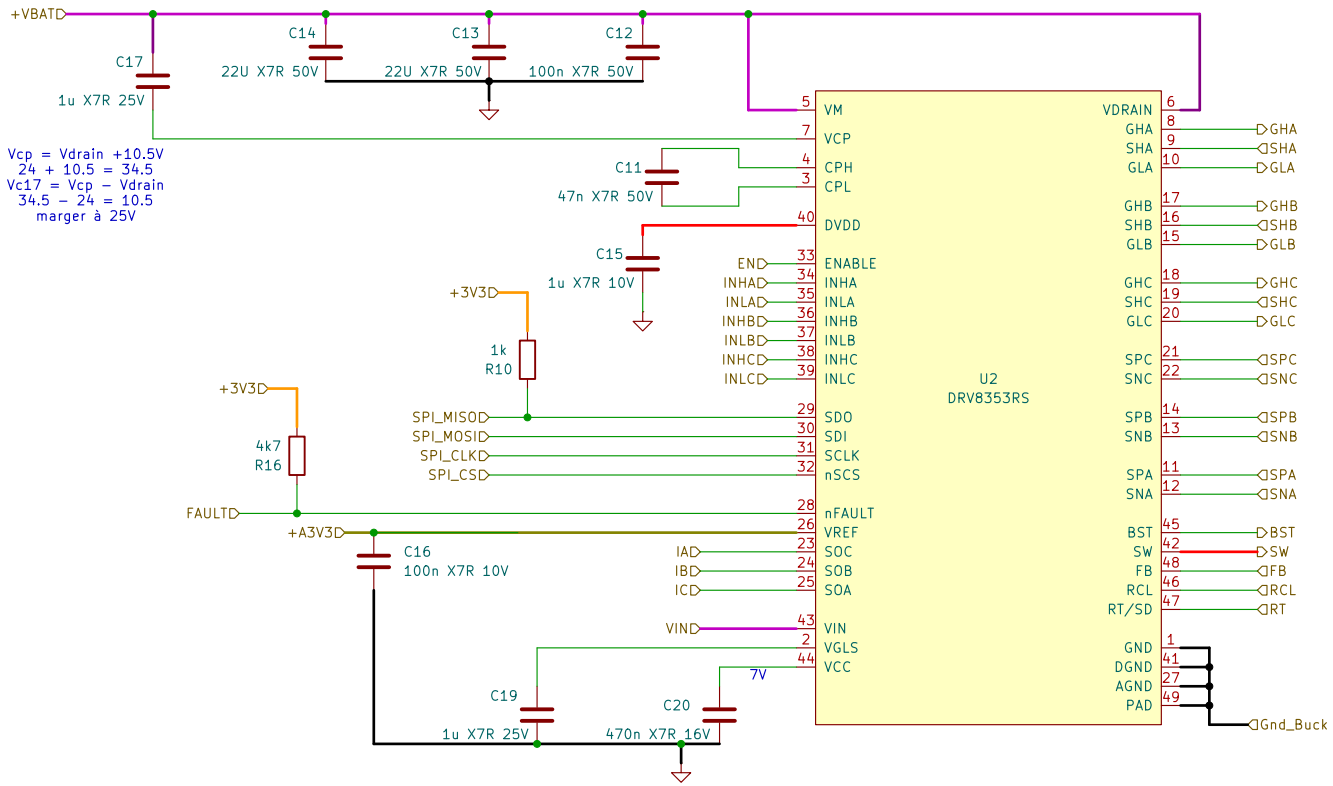
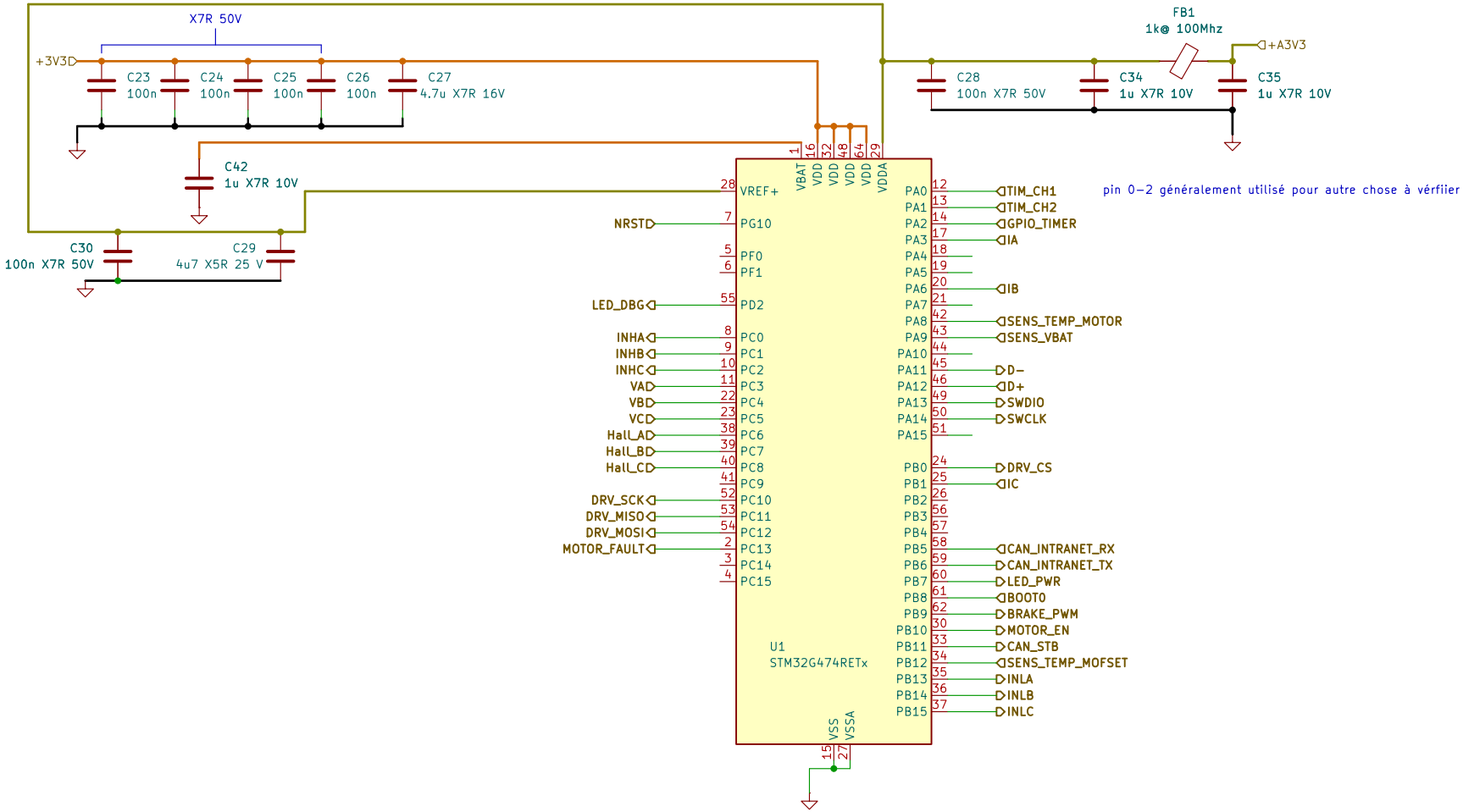
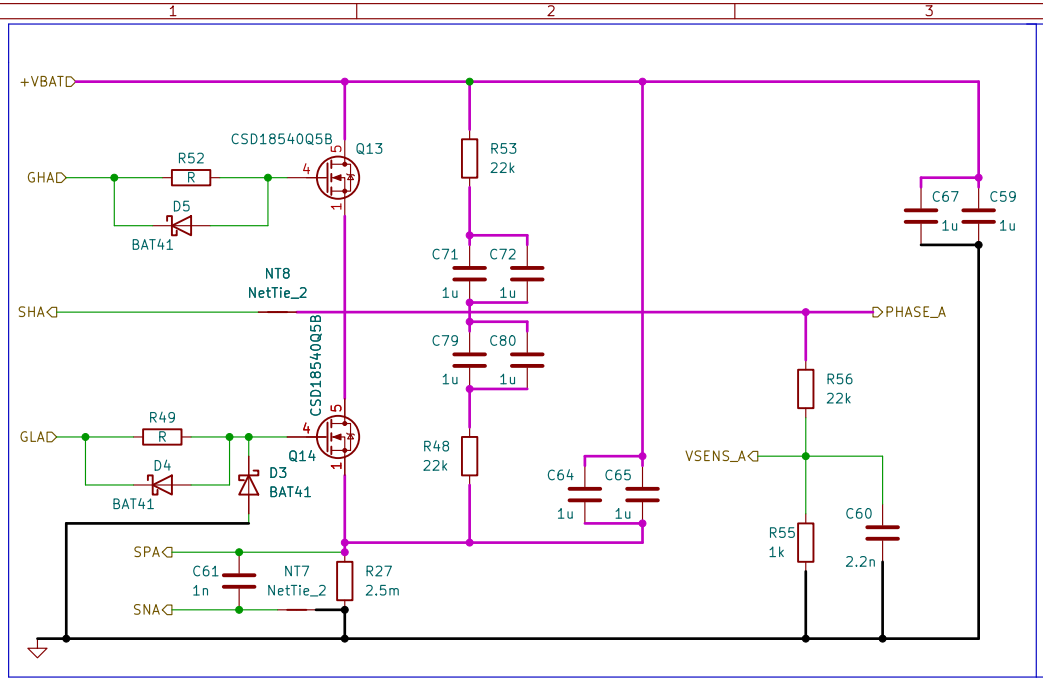

MCU
+5V
+3V3
+A3V3
SWCLK
SWDIO
D+
D-
TIM_CH1
TIM_CH2
GPIO_TIMER
Hall_A
Hall_B
Hall_C
NRST
BOOT0
IA
IB
IC
MOTOR_FAULT
MOTOR_EN
INHAD
INHBD
INHCD
INLAD
INLBD
INLCD
DRV_CSD
DRV_MOSI
DRV_MISO
DRV_SCK
VA
VB
VC
CAN_STBD
CAN_INTRANET_RX
CAN_INTRANET_TX
LED_DBGD
LED_PWRD
SENS_TEMP_MOFSET
SENS_TEMP_MOTOR
SENS_VBAT
BRAKE_PWM
Fichier: MCU.kicad_sch
Connecting
+5V
+3V3
PHASE_A
PHASE_B
PHASE_C
+VBATD
D+
D-
TIM_CH1
TIM_CH2
GPIO_TIMER
Hall_A
Hall_B
Hall_C
NRST
BOOT0
Fichier: motor.kicad_sch
Power_stage
+5V
+3V3
+A3V3
+VBAT
VIN
SW
BST
FB
RT
RCL
Gnd_Buck
Fichier: Power_stage.kicad_sch
driver
+VBAT
+3V3
+A3V3
VIN
SW
BSTD
FB
RT
RCL
Gnd_Buck
IA
IB
IC
FAULT
EN
INHA
INHB
INHC
INLA
INLB
INLC
SPI_CS
SPI_MOSI
SPI_MISO
SPI_CLK
GHAD
GHBD
GHCD
GLAD
GLBD
GLCD
SHA
SHB
SHC
SNA
SNB
SNC
SPA
SPB
SPC
Fichier: driver.kicad_sch
Inverter
+VBAT
PHASE_A
PHASE_B
PHASE_C
GHA
GHB
GHC
GLA
GLB
GLC
SHA
SHB
SHC
SNA
SNB
SNC
SPA
SPB
SPC
VSENS_A
VSENS_B
VSENS_C
Fichier: Inverter.kicad_sch
Sensing
LED_DBG
LED_PWR
SENSE_TEMP_FET
SENSE_TEMP_MOT
SENSE_VBAT
VIN
+A3VA
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Rev:
KiCad E.D.A. 10.0.6
Id: 1/7

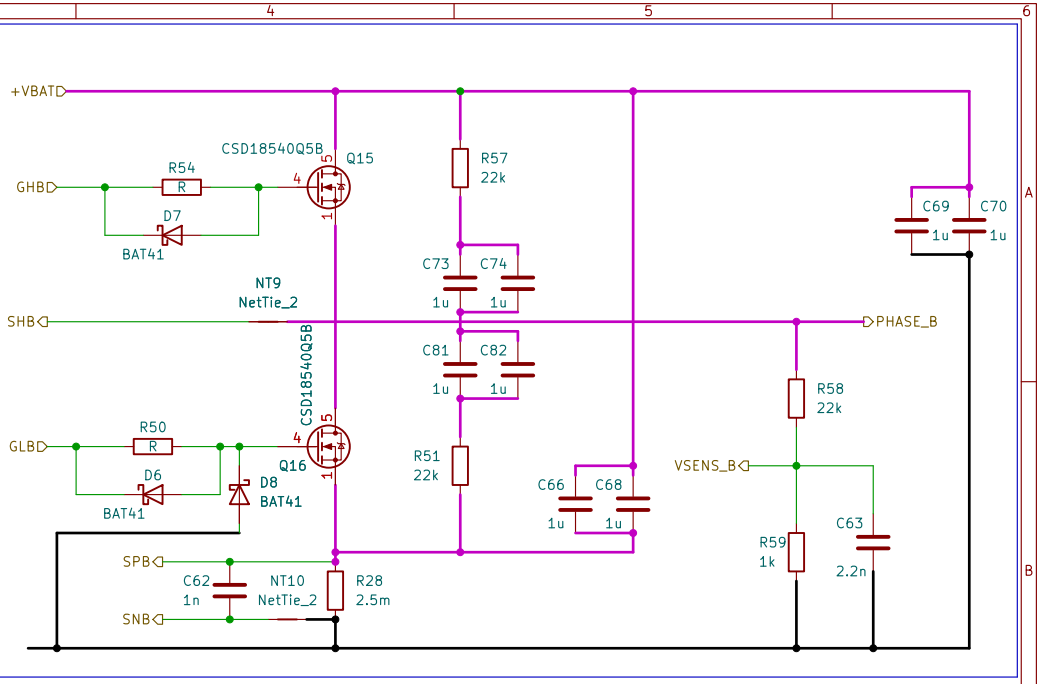


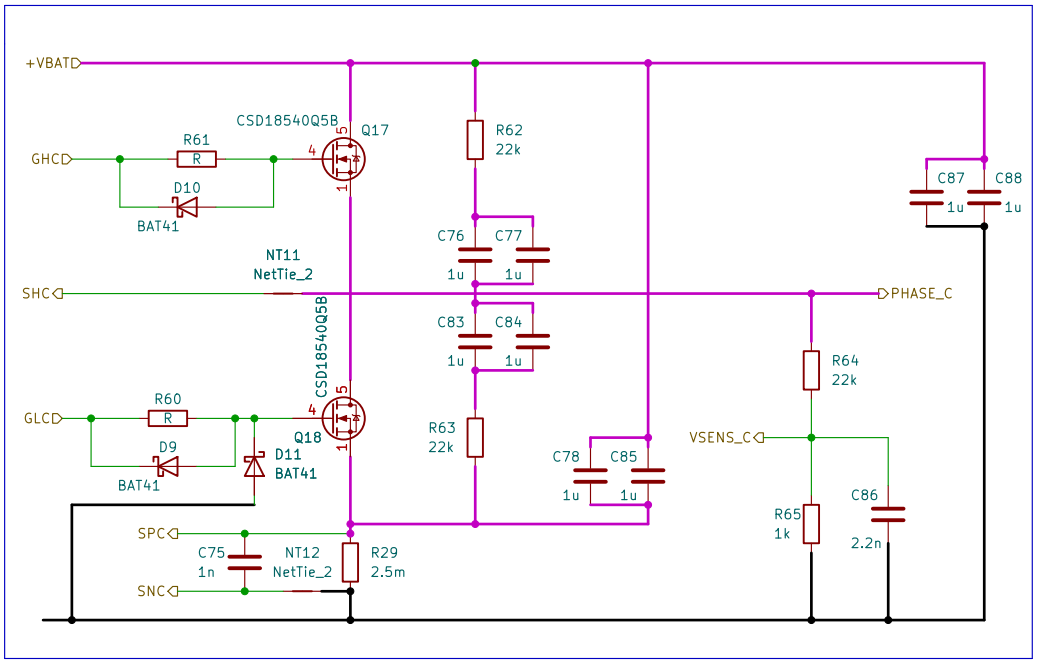
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KiCad E.D.A. 10.0.6		Id: 2/7



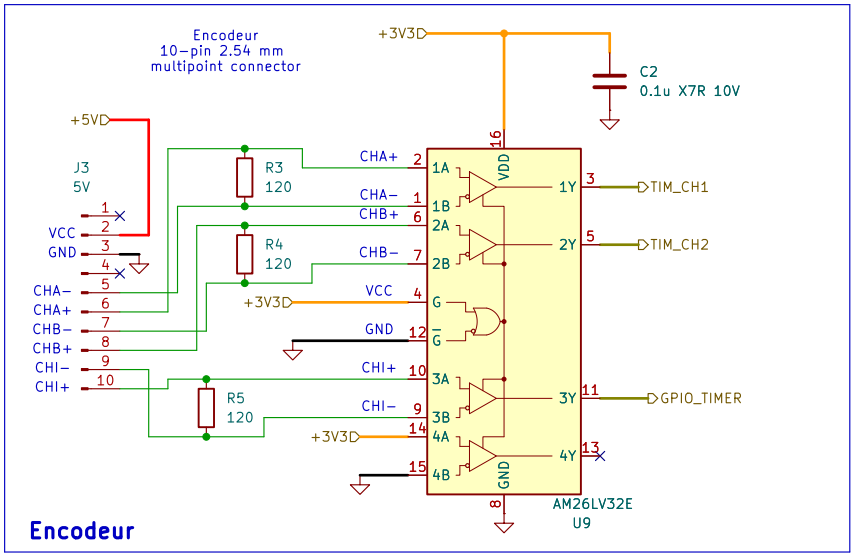
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KiCad E.D.A. 10.0.6		Id: 3/7

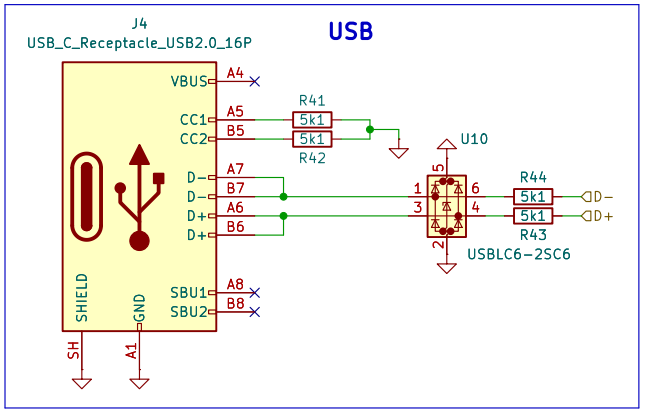


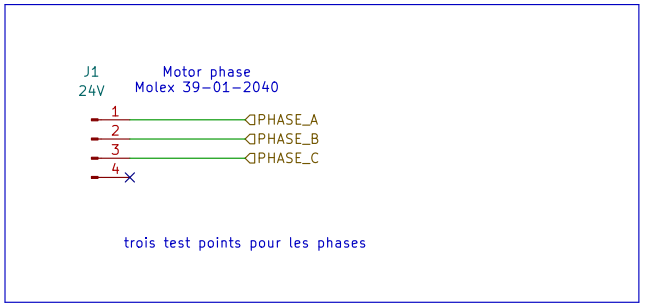


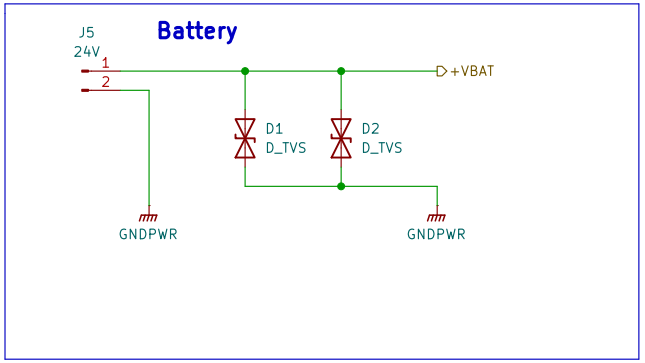


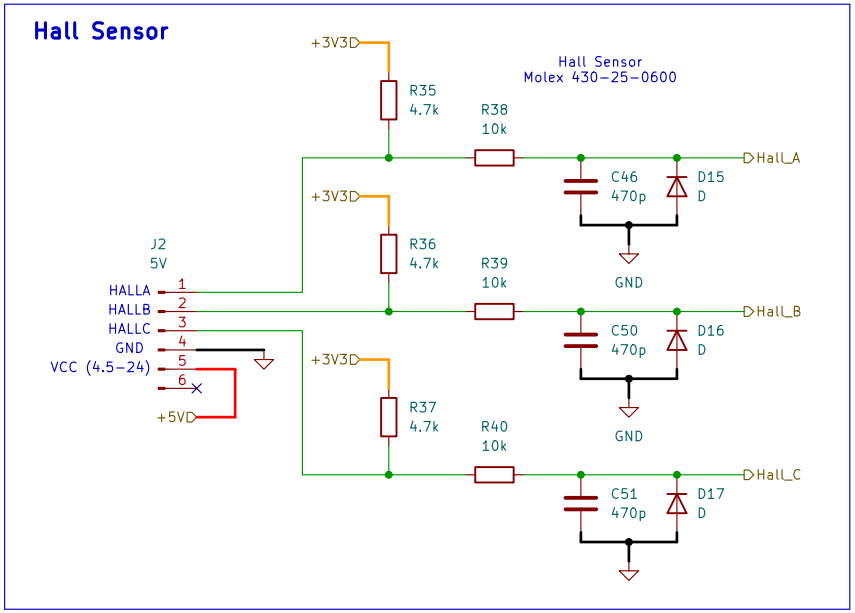
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KiCad E.D.A. 10.0.6		Id: 4/7



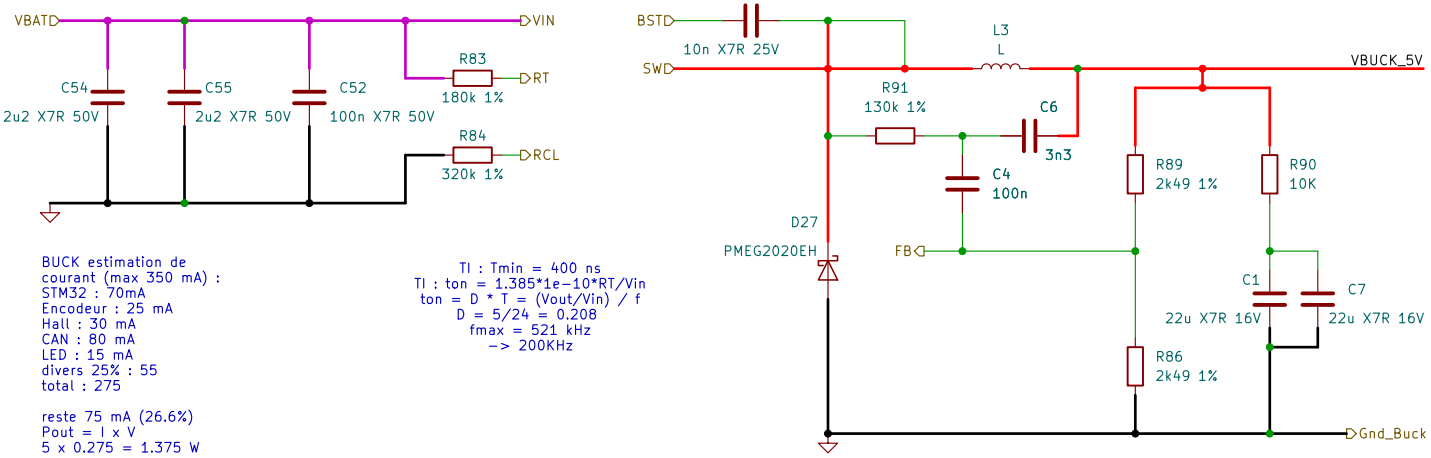








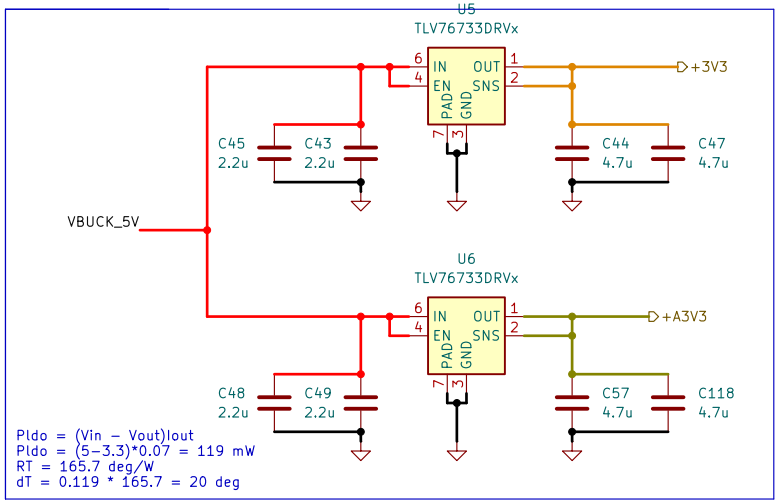
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/7



BUCK estimation de
courant (max 350 mA) :
STM32 : 70mA
Encodeur : 25 mA
Hall : 30 mA
CAN : 80 mA
LED : 15 mA
divers 25% : 55
total : 275

reste 75 mA (26.6%)
Pout = I x V
5 x 0.275 = 1.375 W

TI : Tmin = 400 ns
TI : ton = 1.385*1e-10*RT/Vin
ton = D * T = (Vout/Vin) / f
D = 5/24 = 0.208
fmax = 521 kHz
-> 200KHz

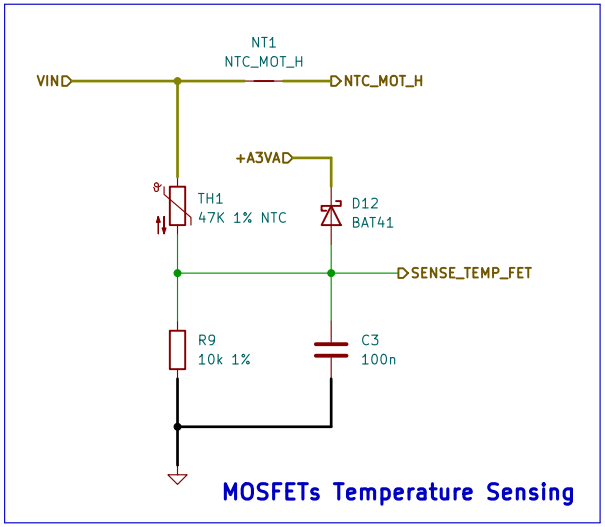


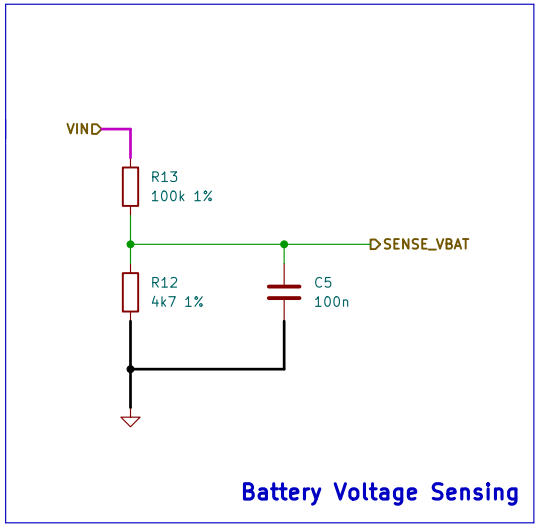
Pldo = (Vin - Vout)Iout
Pldo = (5-3.3)*0.07 = 119 mW
RT = 165.7 deg/W
dT = 0.119 * 165.7 = 20 deg

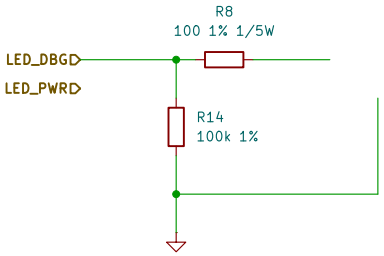
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/7







Sheet: /Sensing/ File: Sensing.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/7